ME5311 Project Report: POD-Based Comparison of Hankel-DMD, SINDy, and NVAR

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This study compares Hankel-extended Dynamic Mode Decomposition (Hankel-DMD), Sparse Identification of Nonlinear Dynamics (SINDy), and Nonlinear Vector AutoRegression (NVAR) for forecasting a high-dimensional spatio-temporal velocity field. Using $15{,}000$ snapshots on a 64×64 grid, Proper Orthogonal Decomposition reduces the $8{,}192$-dimensional state to $r = 31$ modes (92.26% training energy). At a short horizon ($h = 20$), all methods achieve $R^2 > 0.97$. At $h = 200$, SINDy performs best ($R^2 = 0.656$), followed by Hankel-DMD ($R^2 = 0.572$) and NVAR ($R^2 = 0.464$). Hankel-DMD gives the best long-horizon accuracy ($R^2 = 0.150$ at $h = 2000$) and longest valid prediction time ($\mathrm{VPT} = 5.6$ t.u.) but has the highest training cost. SINDy is the fastest (< 0.01 s) and most competitive at medium horizons. All methods eventually diverge at long horizons, consistent with chaotic dynamics.

Dynamic Mode Decomposition | SINDy | NVAR | POD | Spatio-temporal prediction

Introduction

Forecasting the future states of high-dimensional spatio-temporal systems from data is a central problem in computational mechanics and fluid dynamics (1). Within the inverse-problem framework of (2), the objective is to recover the dynamical operator $\mathbf{m}$ satisfying $\mathbf{d} = g(\mathbf{m})$, where $\mathbf{d}$ denotes observed future states and g is the forward propagation map. Recovering $\mathbf{m}$ from a measured trajectory and then using $g(\mathbf{m})$ for prediction is precisely the inverse task.

Three paradigms span the interpretability–flexibility spectrum (2). Dynamic Mode Decomposition (DMD) (3, 4) solves the linearised inverse problem for a best-fit linear propagator; extending DMD with Hankel delay-coordinate embedding (5, 6) expands the effective state space and allows the linear model to capture a broader class of nonlinear dynamics via the Koopman operator framework (7, 8). SINDy (9) identifies a parsimonious governing equation from a polynomial candidate library via sparse regression (10), connecting directly to the ℓ_p-norm cost-function formulations discussed in the course. Nonlinear Vector AutoRegression (NVAR) (11) combines delay-coordinate embedding with nonlinear polynomial features and a closed-form ridge-regression readout, offering a computationally efficient reservoir-computing-inspired alternative to gradient-based neural networks (12).

Direct application of these methods to the raw $\mathcal{O}(10^4)$-dimensional snapshots is computationally intractable. Proper Orthogonal Decomposition (POD) (13), implemented via the snapshot method of Sirovich (14), first projects the data onto an energetically dominant low-dimensional subspace, reducing the prediction problem to one over a compact set of temporal coefficients. The combination of POD with data-driven dynamical models has proven effective for reduced-order modelling of complex flows (2, 15). This paper implements and compares all three approaches on a controlled benchmark, evaluates multi-horizon prediction accuracy, and discusses the trade-offs among accuracy, computational cost, and interpretability.

Methods

Data and dimensionality reduction. The dataset contains $N_t = 15{,}000$ snapshots of a two-component planar velocity field on a uniform 64×64 Cartesian grid with periodic boundary conditions ($\Delta t = 0.2$, $T = 3000$). The data are split *chronologically* into training (60%), validation (20%), and test (20%) sets to prevent temporal data leakage. Each snapshot is vectorised to $\mathbf{x}(t) \in \mathbb{R}^n$, $n = 8{,}192$. The mean-centred training snapshot matrix $\mathbf{X} \in \mathbb{R}^{n \times N_{\text{train}}}$ is factorised as $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$; the leading $r = 31$ left singular vectors form the POD basis $\mathbf{U}_r$, capturing 92.26% of the training energy (Fig. 1a). The temporal coefficients $\mathbf{a}(t) = \mathbf{U}_r^\top \mathbf{x}(t) \in \mathbb{R}^{31}$ serve as the state vector for all downstream models.

Hankel-Extended Dynamic Mode Decomposition. To expand the effective state space, a delay-coordinate embedding of depth d is formed:

$$\mathbf{z}(t) = \left[\mathbf{a}(t)^\top, \mathbf{a}(t-1)^\top, \ldots, \mathbf{a}(t-d+1)^\top\right]^\top \in \mathbb{R}^{rd}, \qquad [1]$$

yielding Hankel snapshot matrices $\mathbf{Z}$ and $\mathbf{Z}'$ (shifted by one step). DMD then identifies the linear map $\mathbf{Z}' \approx \mathbf{A}\,\mathbf{Z}$, solved via the reduced propagator $\widetilde{\mathbf{A}} = \mathbf{U}_r^\top \mathbf{Z}' \mathbf{V} \mathbf{\Sigma}^{-1}$ (4, 5). The delay depth $d \in \{12, 16, 20\}$ and eigenvalue damping $\rho \in \{0.995, 0.999\}$ are selected on the validation set; the best configuration ($d = 20$, $\rho = 0.999$) gives a state dimension of $rd = 620$. Eigenvalues

with magnitude exceeding unity are stabilised to $|\lambda| \to \rho$. Predictions are generated by iterating the Hankel propagator and extracting the leading r components of $\mathbf{z}$.

Sparse Identification of Nonlinear Dynamics. SINDy (9) posits that the POD dynamics satisfy $\dot{\mathbf{a}} = \boldsymbol{\Xi}\,\boldsymbol{\Theta}(\mathbf{a})^\top$, where $\boldsymbol{\Theta} \in \mathbb{R}^{N\times p}$ is a polynomial candidate library and $\boldsymbol{\Xi}$ is a sparse coefficient matrix. Because higher POD modes (indices 5–31) carry $< 1\%$ energy each and evolve nearly stochastically, SINDy is applied only to the $r_s = 4$ energetically dominant modes, retaining 78.5% of training energy. The identification problem is formulated as

$$\boldsymbol{\Xi}^* = \arg\min_{\boldsymbol{\Xi}} \left\| \dot{\mathbf{A}}_s - \boldsymbol{\Xi}\,\boldsymbol{\Theta}^\top \right\|_F^2 + \alpha \|\boldsymbol{\Xi}\|_F^2, \qquad [2]$$

and solved via the Sequential Thresholded Least Squares algorithm ($\tau = 0.005$, $\alpha = 0.1$), using a degree-1 library. The remaining $r - r_s$ coefficients are held fixed at their initial test values during autoregressive rollout. The prediction mode and amplitude-shrink factor ($\alpha_s = 0.9$) are selected on the validation set.

Nonlinear Vector AutoRegression. Following (11), NVAR forms the same delay-coordinate vector $\mathbf{z}(t) \in \mathbb{R}^{rd}$ as Eq. (1) and maps it through a polynomial feature function $\boldsymbol{\Theta} : \mathbb{R}^{rd} \to \mathbb{R}^p$. The readout matrix is obtained in closed form via ridge regression:

$$\mathbf{W}^* = \arg\min_{\mathbf{W}} \left\| \mathbf{Y} - \boldsymbol{\Theta}(\mathbf{Z})\mathbf{W} \right\|_F^2 + \alpha \|\mathbf{W}\|_F^2, \quad \alpha = 10^{-2}, \qquad [3]$$

where $\mathbf{Y}$ are the one-step-ahead targets. Hyperparameters (delay $d \in \{2, 4, 8\}$, polynomial degree, ridge α) are selected on the validation set; the best configuration ($d = 8$, degree 1, $\alpha = 10^{-2}$) renders NVAR a regularised linear regression in the $rd = 248$-dimensional delay space. Predictions are generated autoregressively.

Evaluation metrics. Prediction quality is assessed at multiple forecast horizons h using the following metrics: the coefficient of determination R^2; root mean squared error (RMSE) and its normalised variant (NRMSE $= \mathrm{RMSE}/\sigma_{\mathrm{true}}$); Pearson correlation; and Valid Prediction Time (VPT), defined as the first horizon at which NRMSE exceeds 0.3.

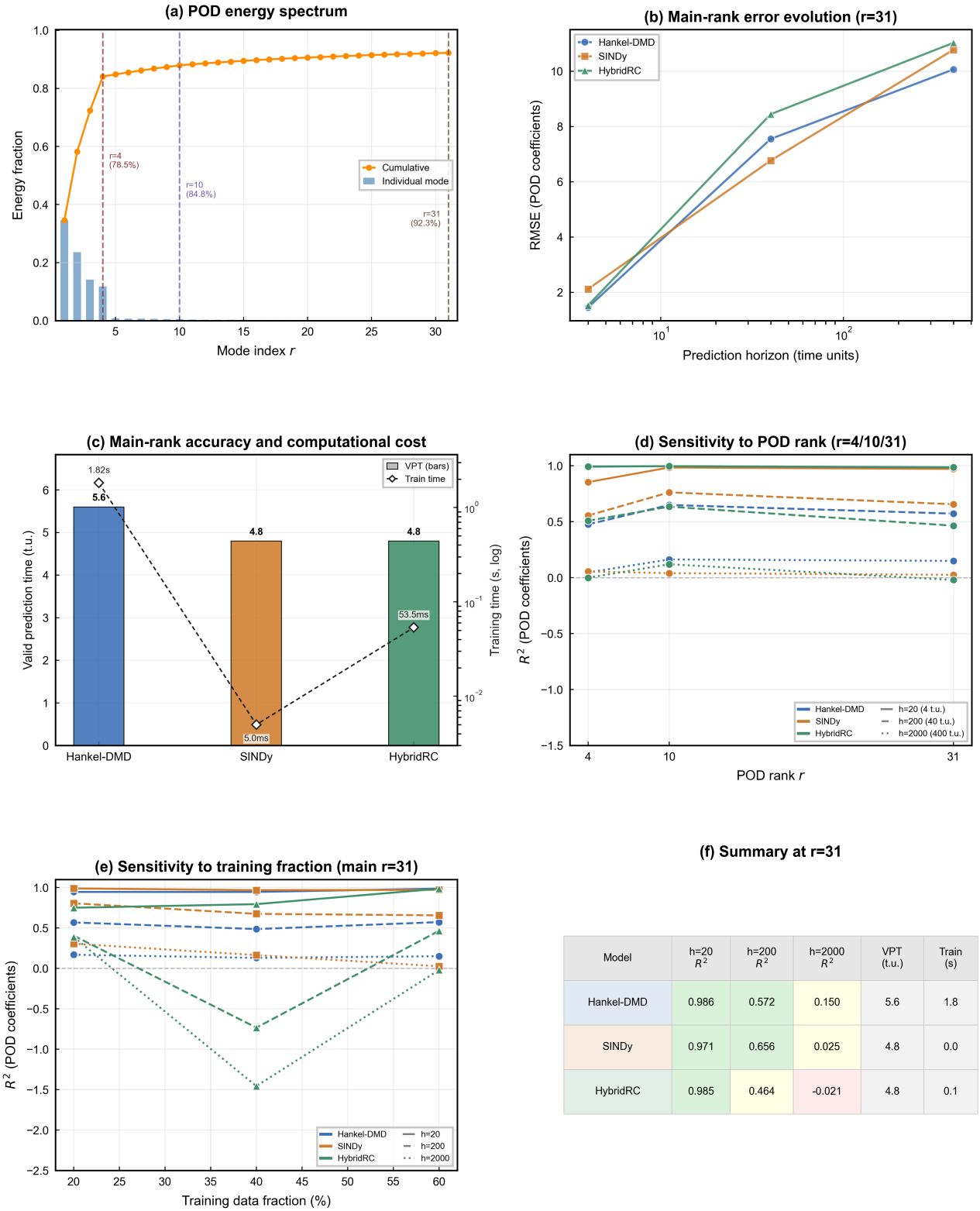
Results and Discussion

Figure 1 summarises the prediction results across all methods. The POD energy spectrum (panel a) reveals a sharp decline from mode 4 (11.8%) to mode 5 (0.73%), consistent with four energetically dominant mode pairs imposed by the prescribed external forcing wavenumber; the chosen rank $r = 31$ retains these dominant modes together with all weaker residual modes up to the 92% energy threshold.

Prediction accuracy. At $h = 20$ steps ($t = 4$ t.u.), all three methods achieve $R^2 > 0.97$ and correlation > 0.99 (panels c, f), with Hankel-DMD performing best ($R^2 = 0.986$), followed by NVAR ($R^2 = 0.985$) and SINDy ($R^2 = 0.971$). Performance differences become pronounced at longer horizons: at $h = 200$ ($t = 40$ t.u.), SINDy leads with $R^2 = 0.656$, followed by Hankel-DMD ($R^2 = 0.572$) and NVAR ($R^2 = 0.464$). SINDy's medium-horizon advantage arises because operating on the $r_s = 4$ dominant modes avoids contamination from the small-scale stochastic fluctuations present in modes 5–31, which degrade the rollout of full-state models. At $h = 2000$ ($t = 400$ t.u.), Hankel-DMD maintains $R^2 = 0.150$ while SINDy degrades to $R^2 = 0.025$ and NVAR to $R^2 = -0.021$. Hankel-DMD also achieves the longest valid prediction time (VPT $= 5.6$ t.u.), while SINDy and NVAR share VPT $= 4.8$ t.u. All methods eventually diverge at long horizons (panel b), consistent with exponential error growth from positive Lyapunov exponents in chaotic systems (16). Rolling-origin evaluation confirms these trends: at $h = 200$, Hankel-DMD yields $R^2 = 0.447 \pm 0.164$ and NVAR $R^2 = 0.398 \pm 0.197$, while SINDy shows higher variance ($R^2 = 0.270 \pm 0.662$) due to sensitivity to initial conditions when operating in the reduced subspace.

Computational cost and interpretability. SINDy trains in < 0.01 s and yields explicit governing equations for the four dominant modes, making it the most interpretable and fastest method; its medium-horizon accuracy demonstrates that focusing on the energetically dominant subspace is beneficial. NVAR requires $\sim$0.05 s for the closed-form ridge solve, with no iterative optimisation; its degree-1 delay library renders it a linear model in the augmented state space, analogous to Hankel-DMD but without the spectral decomposition step. Hankel-DMD requires $\sim$1.82 s training due to the large state dimension ($rd = 620$), but achieves the best long-horizon accuracy and VPT by exploiting the full delay-coordinate structure to approximate Koopman eigenfunctions (7, 8). Overall, the three methods offer complementary trade-offs: SINDy for interpretability and medium-horizon accuracy, NVAR for computational efficiency, and Hankel-DMD for long-horizon robustness; all are ultimately bounded by the fundamental predictability limit of chaotic dynamics.

AI Declaration. AI-assisted language tools (Claude Sonnet 4.6) were used for limited drafting and wording refinement of the manuscript; all analysis design, code implementation, result verification, and technical interpretation were completed by the author.



Model	h=20 R^2	h=200 R^2	h=2000 R^2	VPT (t.u.)	Train (s)
Hankel-DMD	0.986	0.572	0.150	5.6	1.8
SINDy	0.971	0.656	0.025	4.8	0.0
HybridRC	0.985	0.464	-0.021	4.8	0.1

Fig. 1. Prediction analysis for Hankel-DMD, SINDy, and NVAR. (a) POD energy spectrum on training data; bar: per-mode fraction; curve: cumulative energy; vertical dash-dot: $r = 31$; horizontal dashed: 90% threshold. (b) RMSE of predicted POD coefficients versus forecast horizon. (c) True and predicted trajectories of the leading POD coefficient $a_1(t)$ over the first 200 test steps. (d) Reconstructed velocity-magnitude snapshots at test step $t = 50\Delta t$: reference (top), Hankel-DMD and SINDy predictions (bottom row). (e) Pearson correlation versus forecast horizon. (f) R^2 comparison at forecast horizons $h \in \{20, 200, 2000\}$ steps for all three models.

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